Engineering demonstration and defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Engineering demonstration and defense before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone release review

Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Capstone service implementation

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Security and privacy review

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

State what was mocked and what remains untested.

Practise with fading help

> Complete the supported Engineering demonstration and defense trace and record the first step you can explain without help.

Use W2: Trace the worked case for Engineering demonstration and defense. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Engineering demonstration and defense explanation against the supplied facts before discussing it.

Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Engineering demonstration and defense facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Engineering demonstration and defense facts and keep your reasoning.

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Engineering demonstration and defense facts and keep your reasoning.

A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Engineering demonstration and defense facts and keep your reasoning.

> Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Engineering demonstration and defense facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Engineering demonstration and defense to Grounded Learning Service, identify a limit and choose the next useful test.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback.

What can be claimed? Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Engineering demonstration and defense

> Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Explain the labelled step before checking the changed condition.

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Approved card V7 says club V uses room E. All trusted tools are read-only.

AI ACADEMY / AI-6/36

Engineering demonstration and defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show source path

Connect the question to retrieved card V7.



2. Support the claim

Room E follows from V7, not merely fluent wording.



3. Show authority

The service cannot edit rooms or send messages.



4. Name missing evidence

With no approved card, decline to invent a room.

CHANGE ONE THING

The challenger asks the bot to email everyone about room E.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With no approved card, decline to invent a room.

> 1. Name missing **evidence**

2. Show source path

3. Support the claim

4. Show authority

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Name missing **evidence** With no approved card, decline to invent a room. A defence links behaviour to **evidence** and limitations.

Demonstrate both a valid answer path and a meaningful refused request.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The challenger asks the bot to email everyone about room E. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Refuse that operation while retaining the supported room answer.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Repair a missing building block

Review Capstone release review (ai-6/35). Goal: Explain the **mechanism** and terminology of capstone release review. Short practice: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?. Return to this week and explain what you changed.

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

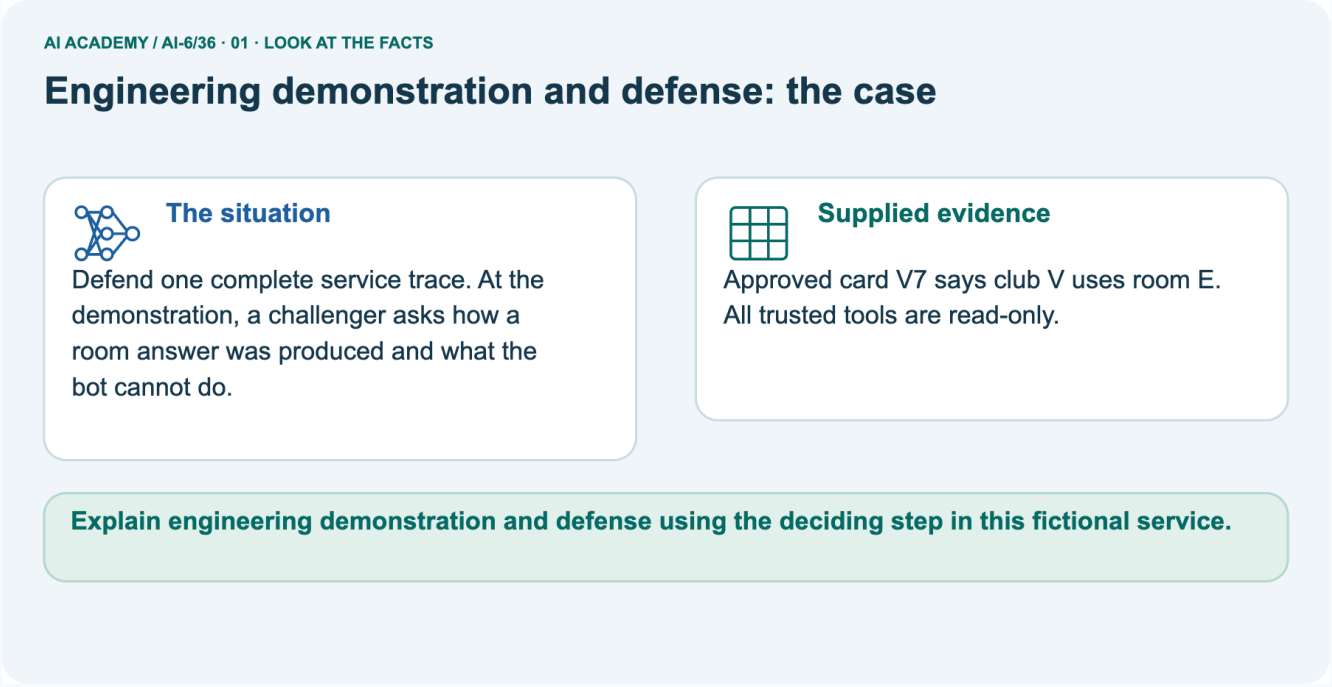
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Engineering demonstration and defense picture from its labels, then record a relevant limit.

01. Engineering demonstration and defense: the case



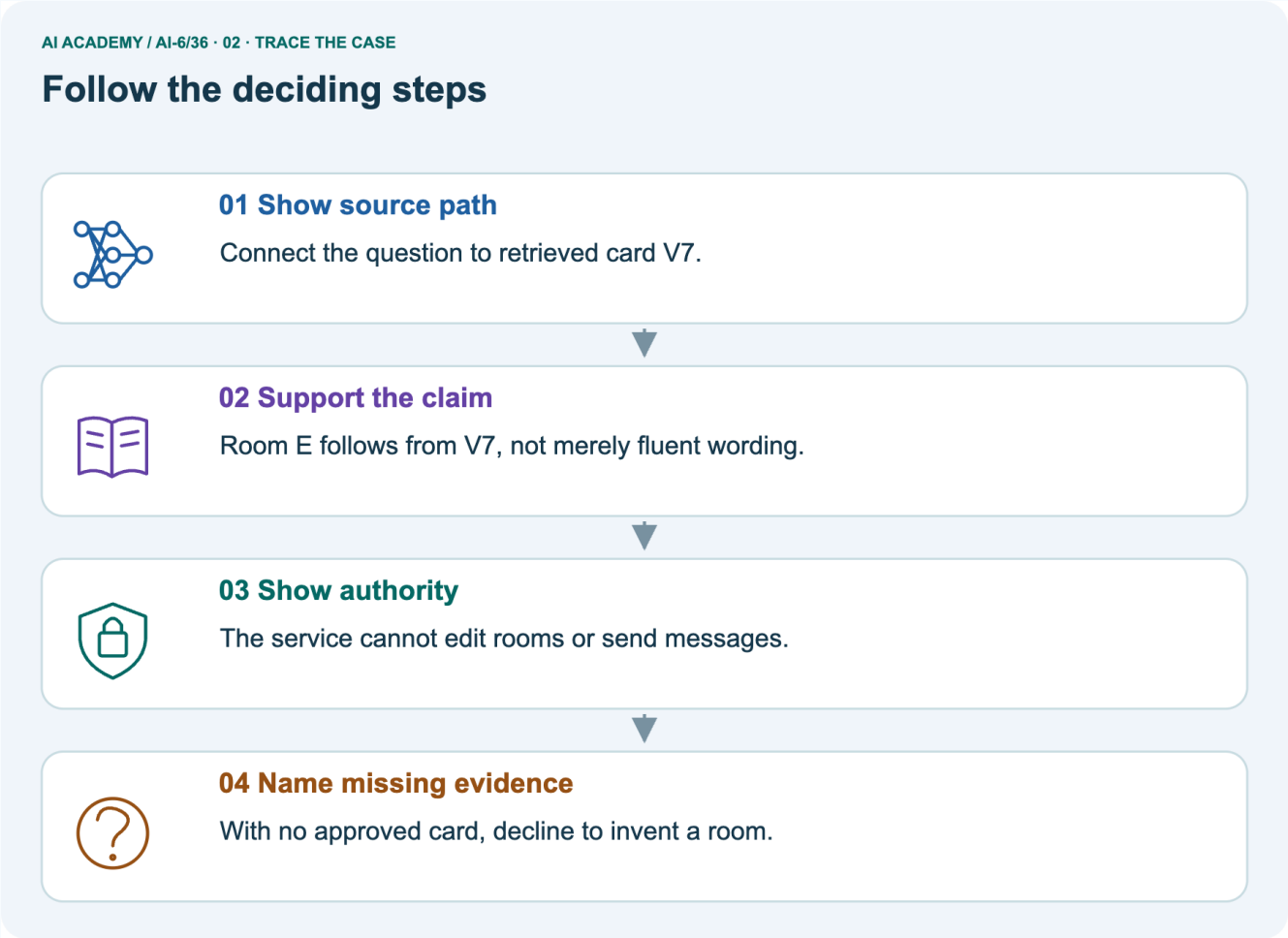
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Supplied facts: Approved card V7 says club V uses room E. All trusted tools are read-only.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



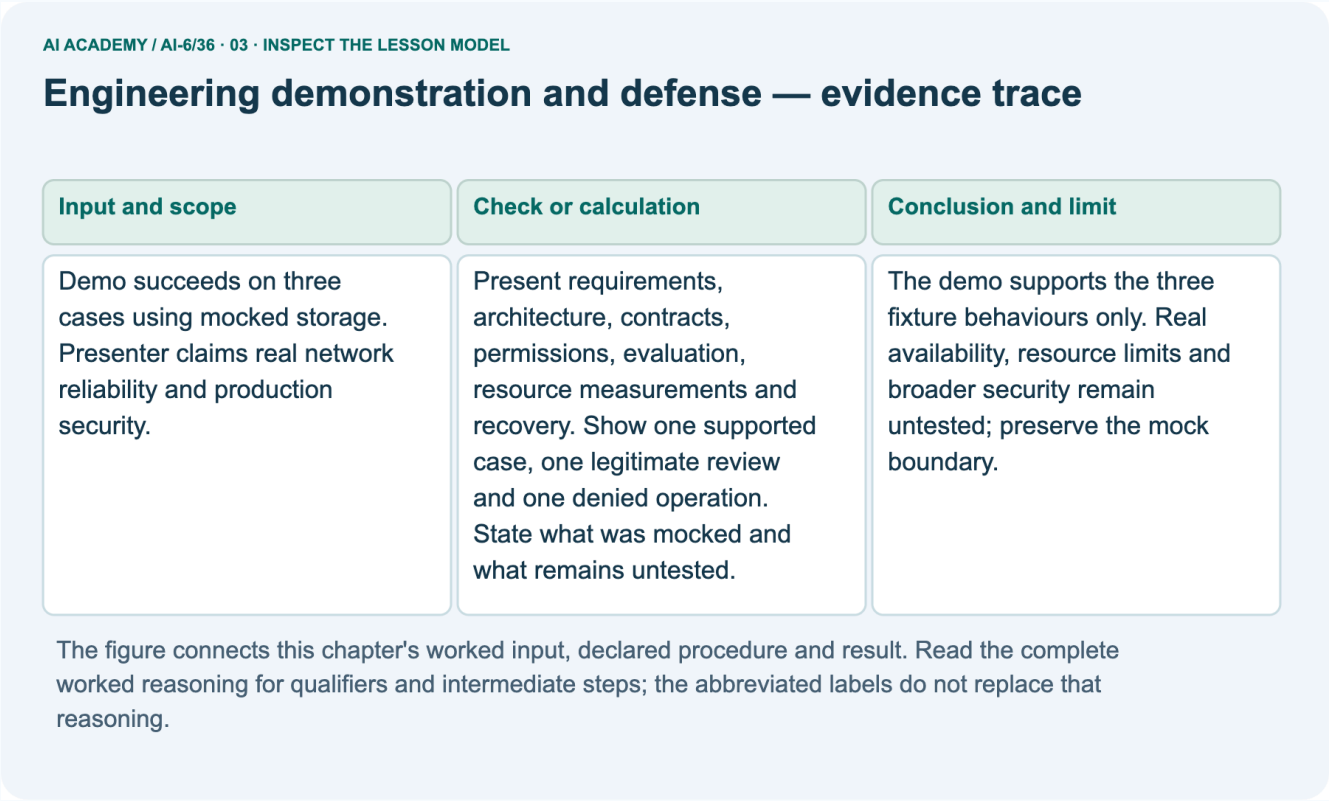
Follow the labelled picture with the matching explanation. Open the vector diagram

Show source path: Connect the question to retrieved card V7. → Support the claim: Room E follows from V7, not merely fluent wording. → Show authority: The service cannot edit rooms or send messages. → Name missing **evidence**: With no approved card, decline to invent a room.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Engineering demonstration and defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineering demonstration and defense — **evidence** trace. Demo succeeds on three cases using mocked storage.

Presenter claims real network reliability and production security. | Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery.

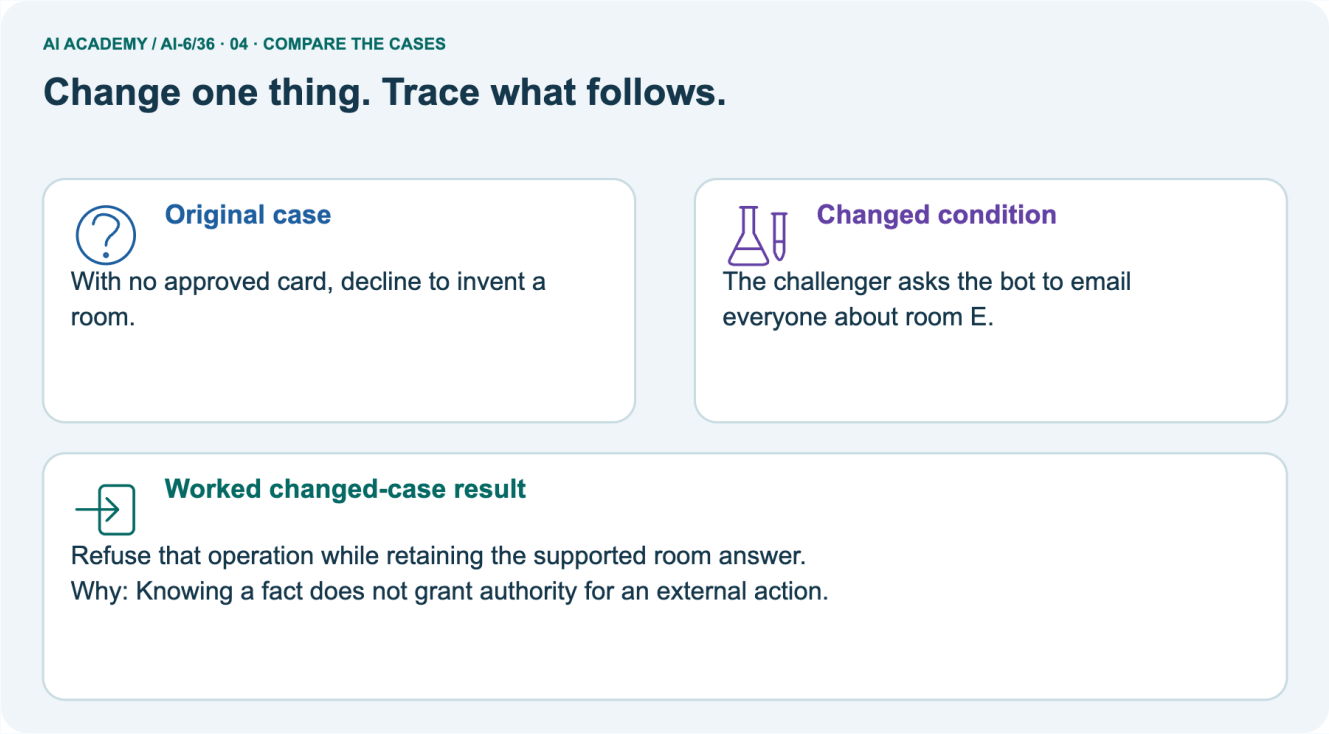
Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested. | The demo supports the three fixture behaviours only.

Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

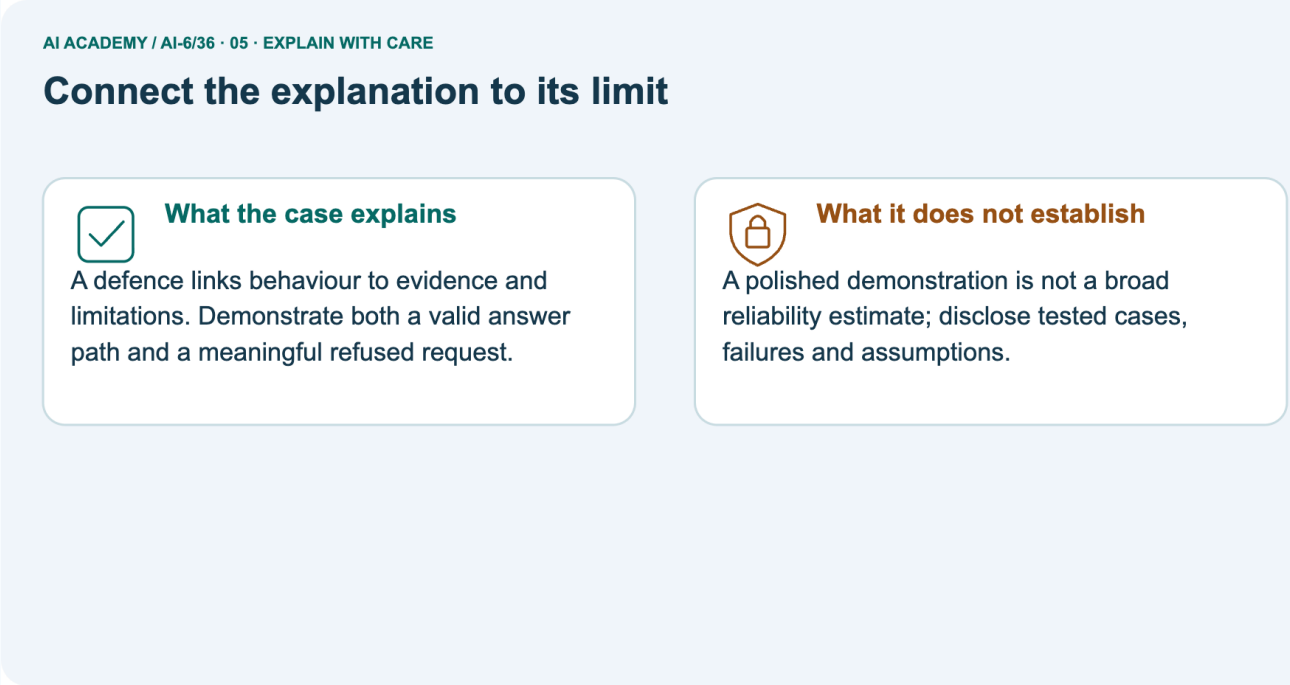
Original case: With no approved card, decline to invent a room. Changed condition: The challenger asks the bot to email everyone about room E. Expected result: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

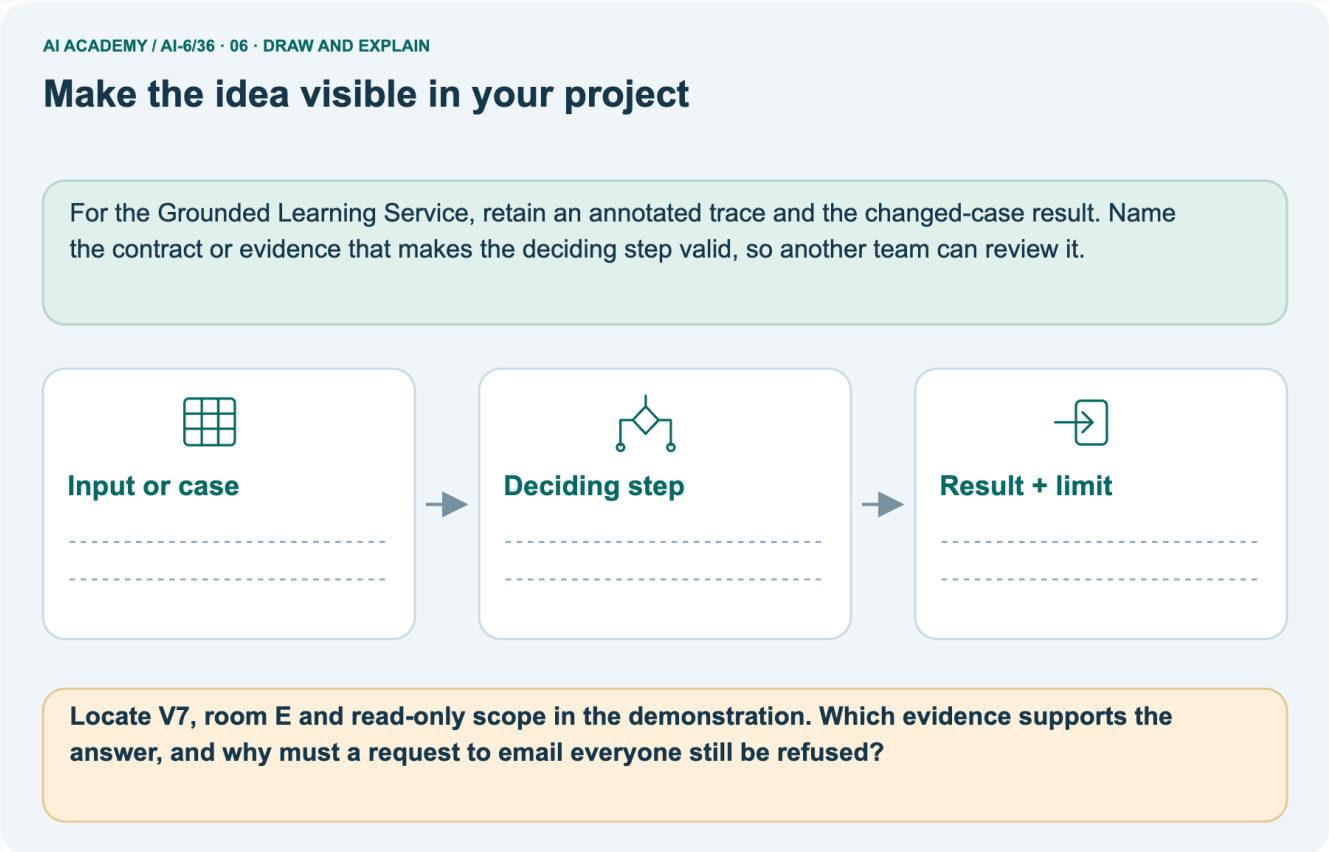
A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Limit: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.